

CS 316: Introduction to Deep Learning

Logistics and Introduction
Week 1

Dr Abdul Samad

Lecture Outline

- Idea of Machine Learning Software
- Key Components of Machine Learning
- Kinds of Machine Learning Problems
- Road to Deep Learning
- Examples of Deep Learning Problems

How to Design Software Solutions.

- **Can we write programs for the following:**
 - To predicts weather given geographic information, satellite images, and a trailing window of past weather.
 - That takes in a question, expressed in free-form text, and answers it correctly.
 - Given an image can identify all the people it contains,
 - drawing outlines around each.
 - Presents users with products that they are likely to enjoy but unlikely, to encounter.

Training Process in Machine Learning

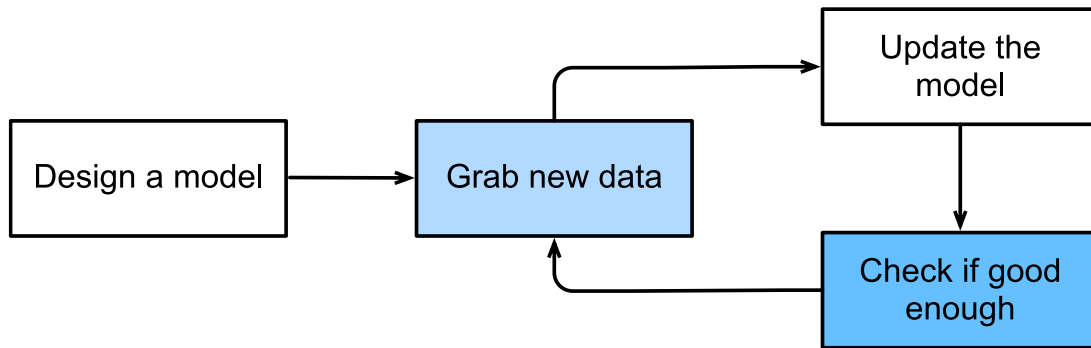


Figure taken from d2l.ai

Key Components of Machine Learning

- Data
- Model
- Objective Function
- Optimization Algorithm

Data

- A collection of examples consists of sets of attributes called features.
- In each example, there is a designated label attribute to predict.
- Features can have a fixed or variable length.
- The quantity of data available determines the effectiveness of deep learning models.
- It is important to ensure high-quality data because the output depends on the input.
- The data can be biased, such as through CV filtering.

Model

- The computational machinery for ingesting data of one type and spitting out predictions of a possibly different type.
- Can be estimated from the data.
- Deep learning models are much more complicated than traditional models.

Objective Function

- To improve our ability to perform a task over time, we need to learn from our experiences.
- Objective function evaluates the quality of our models.
- The lower the objective function value, the better the model is considered to be.
- It is also known as the loss function.
- Mean Squared Error is used for numerical data whereas accuracy is used for categorical data.
- To avoid overfitting, we keep track of both the training and test losses.

Optimization Algorithm

- Given a data source, a model, and a well-defined objective function, we need an algorithm capable of searching for the best possible parameters that minimize the loss function.
- Gradient Descent
- Stochastic Gradient Descent
- Mini Batch Gradient Descent
- Adam

Types of Machine Learning Problems

- Supervised Learning
 - Regression
 - Classification
- Unsupervised Learning

Supervised Learning

- We aim to develop a model that maps input features to labels.
- An example consists of a pair of features and a corresponding label.
- For instance, "heart attack" or "no heart attack" is a label, while vital signs are input features.
- In probabilistic terms, we're interested in estimating the conditional probability of a label based on input features.

Examples of Supervised Learning

- Predict cancer vs. not cancer, given a computer tomography image.
- Predict the correct translation in French, given a sentence in English.
- Predict the price of a stock next month based on this month's financial reporting data.

Supervised Learning System Diagram

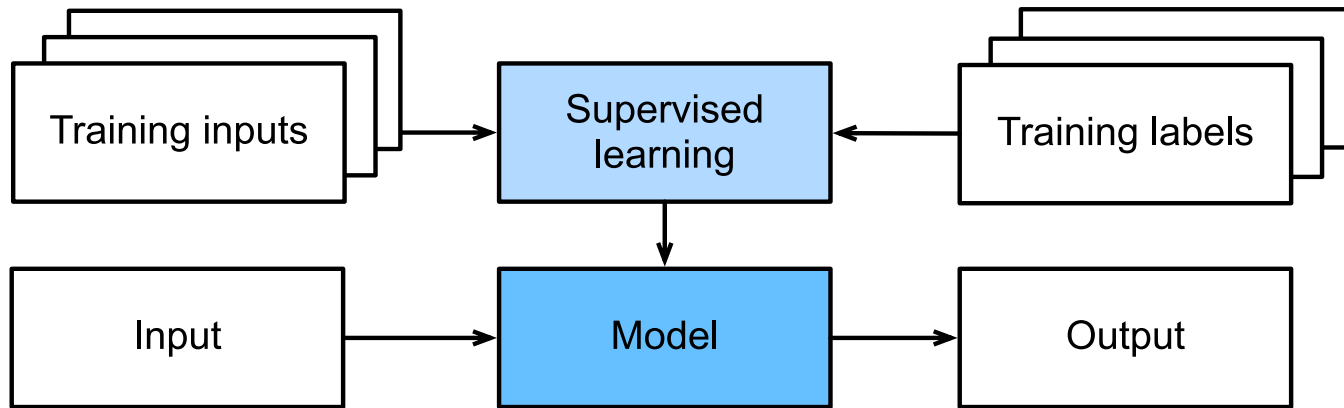


Figure taken from d2l.ai

Supervised Learning - Regression

- Label to predict is a numerical value.
- House Price prediction
- Predict the price of stock next month based on this month's financial reporting data
- How many hours will this surgery take?
- How much rainfall will this town have in the next six hours?

Supervised Learning - Classification

- Label to predict is a category or class.
- Predict whether the image is of a cat or dog.
- Predict whether the email is a spam or not.
- Predict the author of a particular poem.

Unsupervised Learning

- No specific labels have been assigned.
- Can we group a given set of photos into categories such as landscape, dogs, babies, cats, and mountain peaks ?
- Can we group a collection of users' browsing activities based on their similar behavior ?
- Subspace estimation
- Principal Component Analysis
- Generative Adversarial Networks (GANs)

Road to Deep Learning

Decade	Dataset	Memory	Floating point calculations per second
1970	100 (Iris)	1 KB	100 KF (Intel 8080)
1980	1 K (house prices in Boston)	100 KB	1 MF (Intel 80186)
1990	10 K (optical character recognition)	10 MB	10 MF (Intel 80486)
2000	10 M (web pages)	100 MB	1 GF (Intel Core)
2010	10 G (advertising)	1 GB	1 TF (NVIDIA C2050)
2020	1 T (social network)	100 GB	1 PF (NVIDIA DGX-2)

Figure taken from d2l.ai

Classifying Images

Leaderboard

Dataset

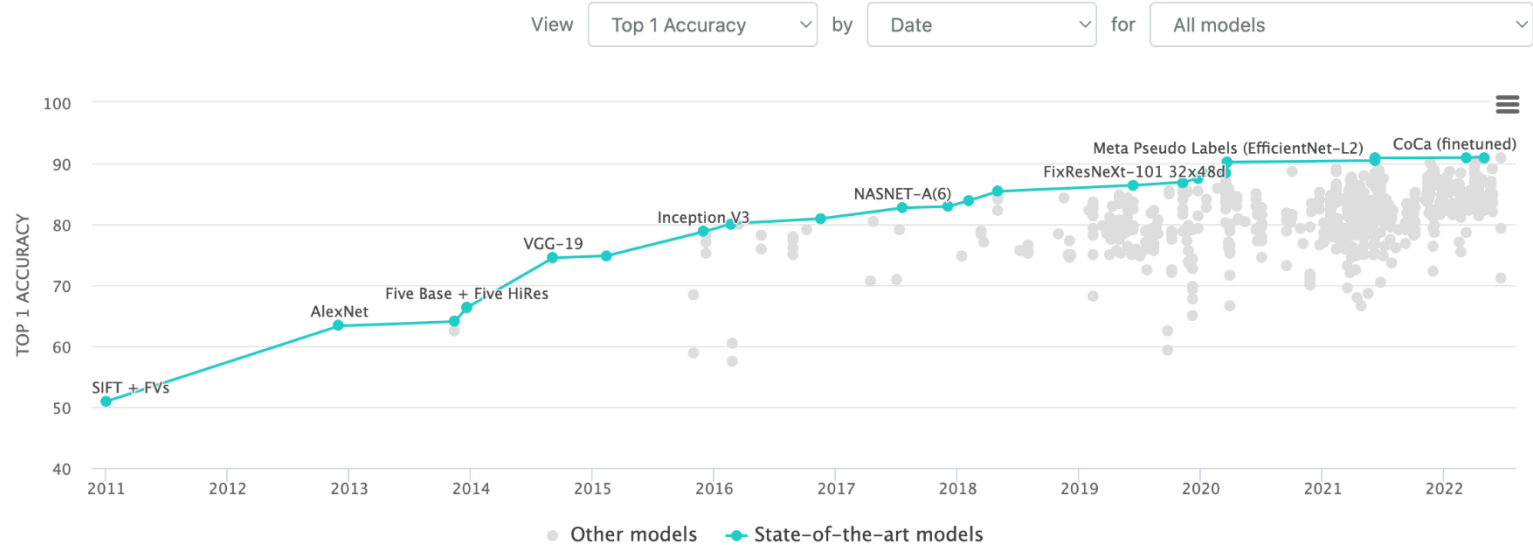


Figure taken from [thematic](#)

Object Detection and Segmentation



Figure taken from [medium](#)

Image Style transfer



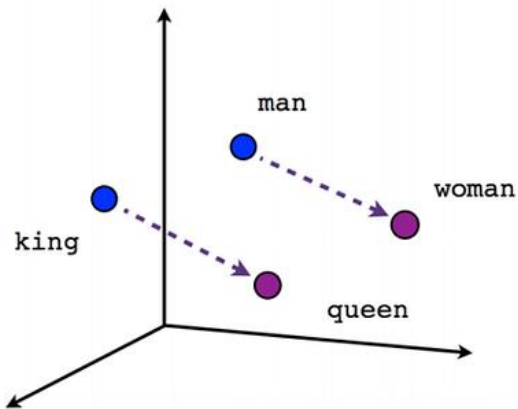
Figure taken from [Analytics Vidhya](#)

Synthesize Faces

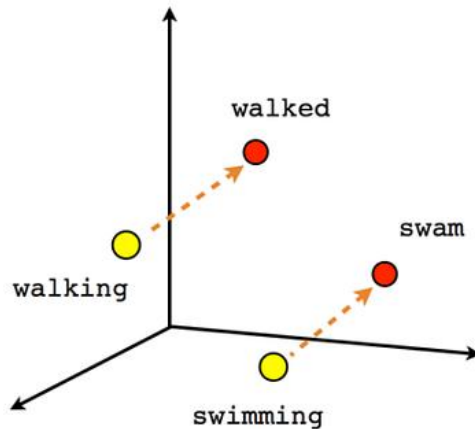


Figure taken from [medium](https://medium.com)

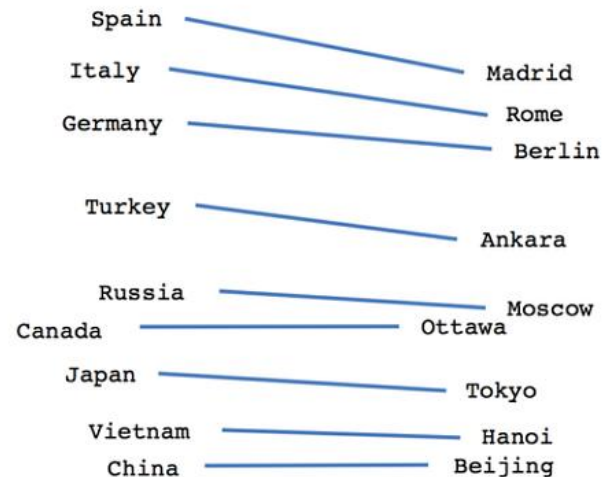
Analogies



Male-Female



Verb tense



Country-Capital

Machine Translation

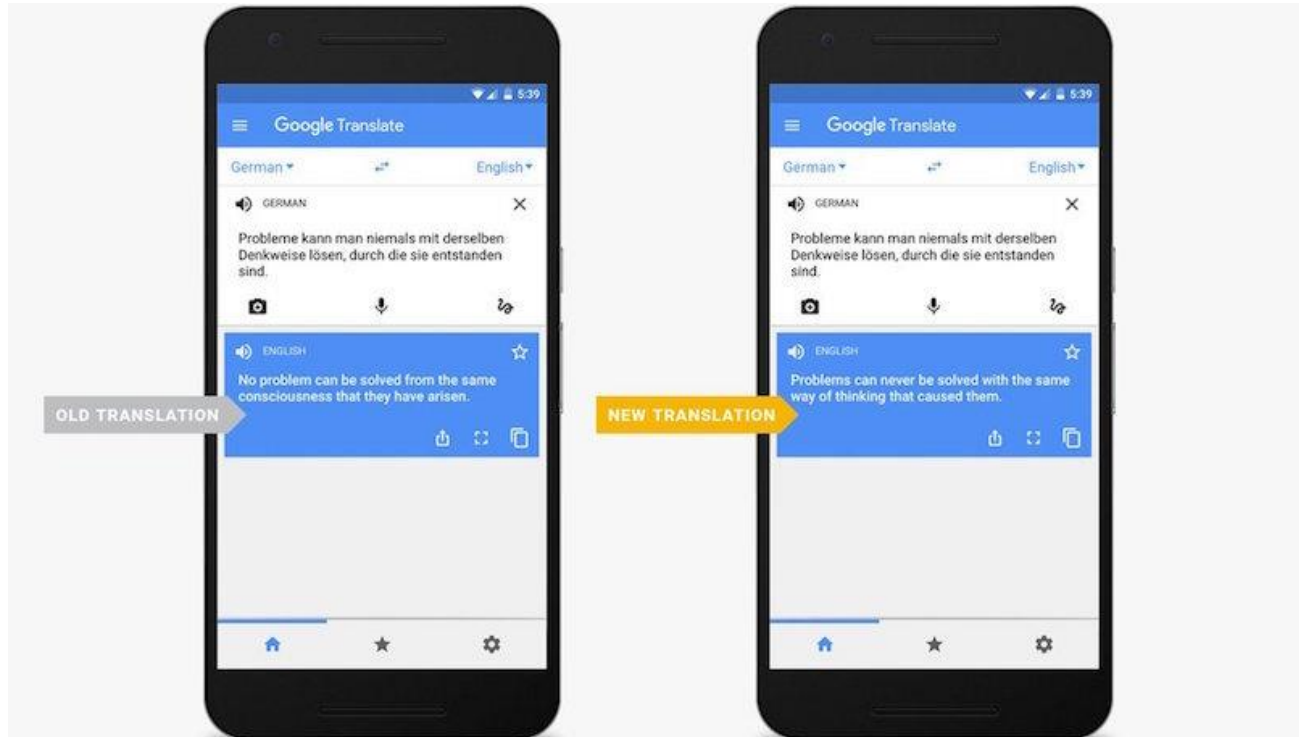


Figure taken from [pcmag](https://www.pcmag.com)

Image Captioning



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."

References

- https://d2l.ai/chapter_introduction/index.html

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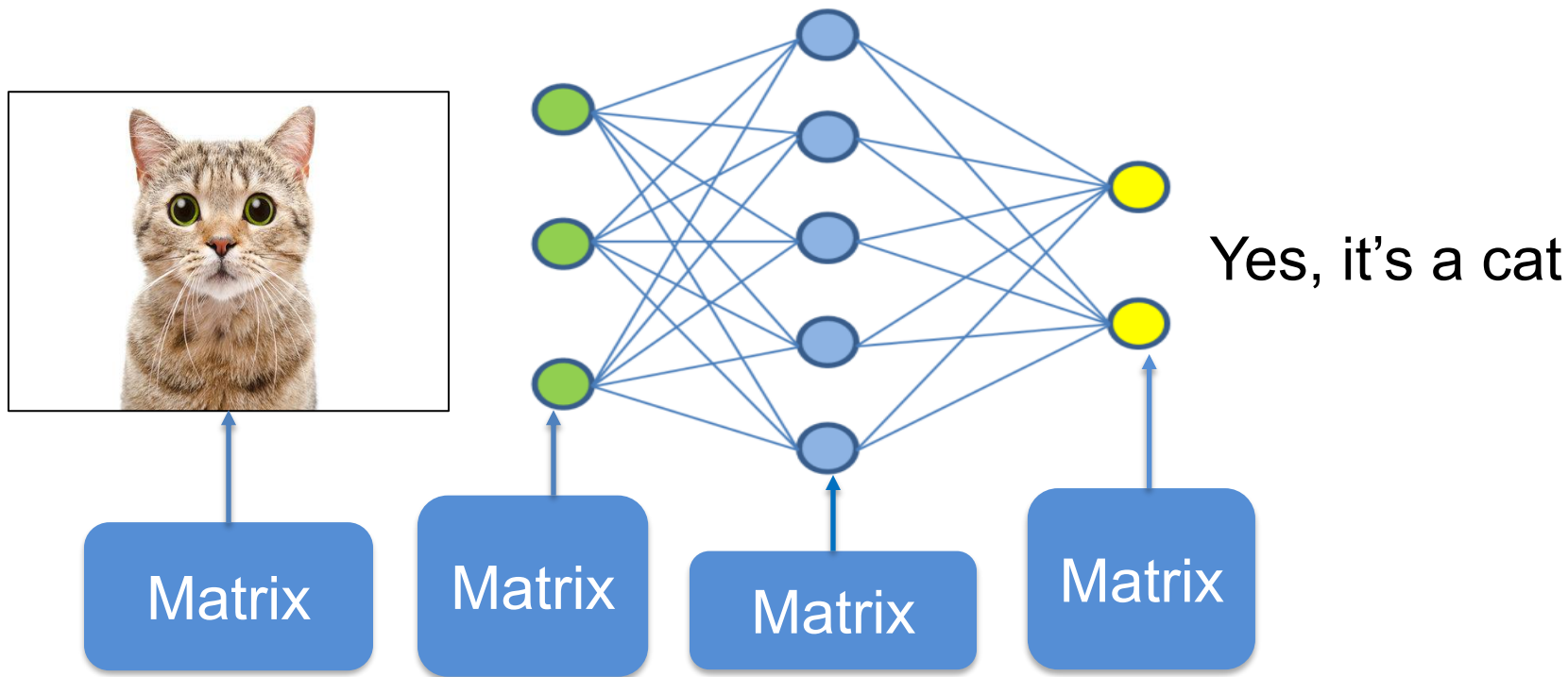
Linear Algebra Overview Week 2

Dr Abdul Samad

Lecture Outline

- Motivation
- Notation
- Scalars
- Vectors
- Matrices
- Vector and matrix calculations

Neural Networks – Matrix Operations



Slide adapted from [DeepLearning.ai](https://www.deeplearning.ai)

Notation

- Scalar – represented by lower case letter (a)
- Vector – represented by bold lower case letter ($\mathbf{a}$)
- Matrix – represented by bold upper case letter ($\mathbf{A}$)

This notation will be used throughout the course.

Scalars vs Vectors vs Matrices vs Tensors

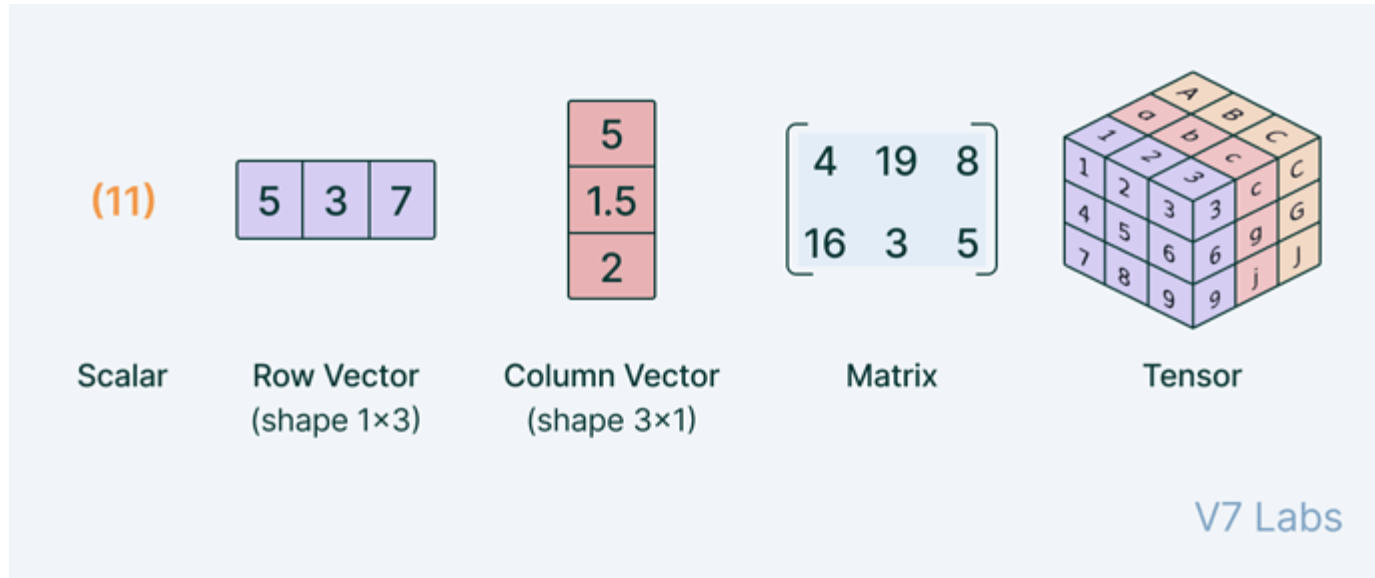


Figure taken from [Kaggle](#)

Scalar

- Variable described by a single number
- 0th order tensor
- Simple Operations
 - Addition
 - Multiplication
 - Function
- Length

Scalar Operations

- Addition

$$c = a + b$$

- Multiplication

$$c = a \cdot b$$

- Function

$$c = \sin(a)$$

Length

- $|a| = \begin{cases} a & \text{if } a > 0 \\ -a & \text{otherwise} \end{cases}$

- $|a + b| \leq |a| + |b|$

- $|a \cdot b| = |a| \cdot |b|$

Vector

- Variable described by a list of scalars
- 1st order tensor
- Simple Operations
 - Addition
 - Multiplication
 - Function
- Length

Vector Orientation

- Vector can be either written as a row vector or a column vector.
- Column Vector

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

- Row Vector

$$\mathbf{x}^T = [1 \ 0 \ 1]$$

Geometric view – Point in Space

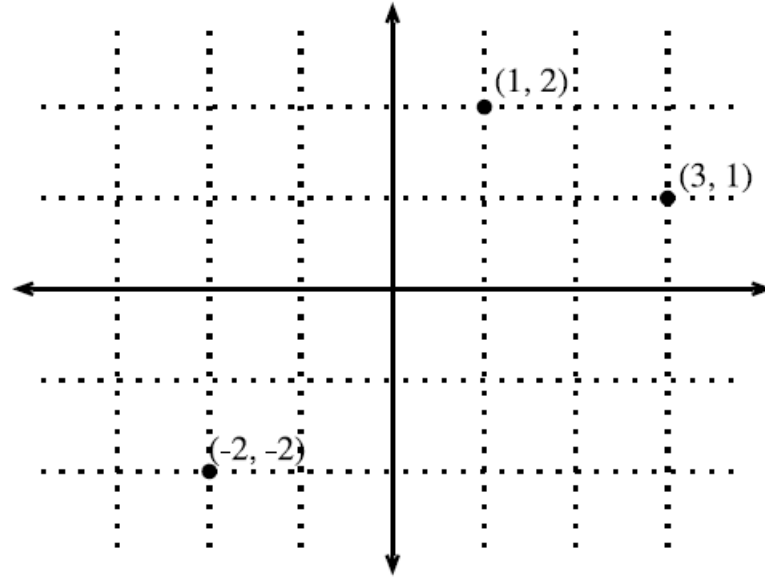


Figure taken from d2l.ai

Geometric view – Direction in Space

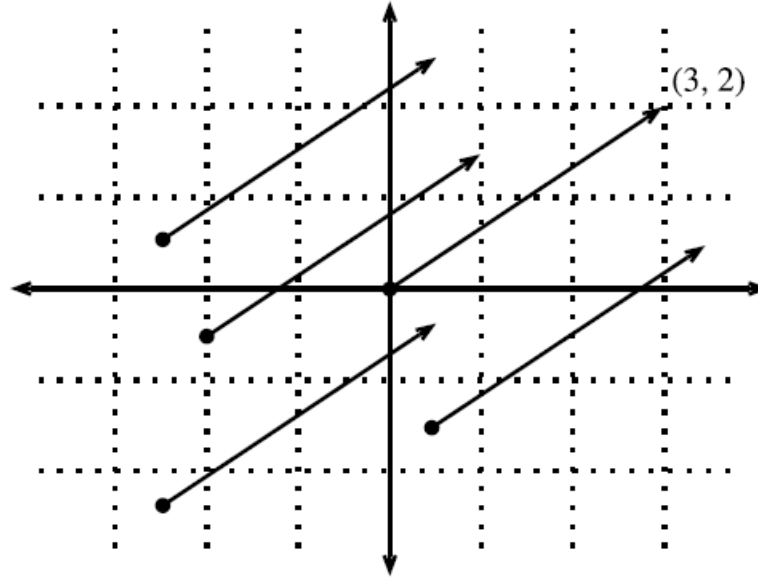


Figure taken from d2l.ai

Vector Operations

- Addition

$$\mathbf{c} = \mathbf{a} + \mathbf{b} \text{ where } c_i = a_i + b_i$$

- Multiplication

$$\mathbf{c} = \mathbf{a} \cdot \mathbf{b} \text{ where } c_i = a_i \cdot b_i$$

- Function

$$\mathbf{c} = \sin(\mathbf{a}) \text{ where } c_i = \sin(a_i)$$

Geometric view - Vector Addition

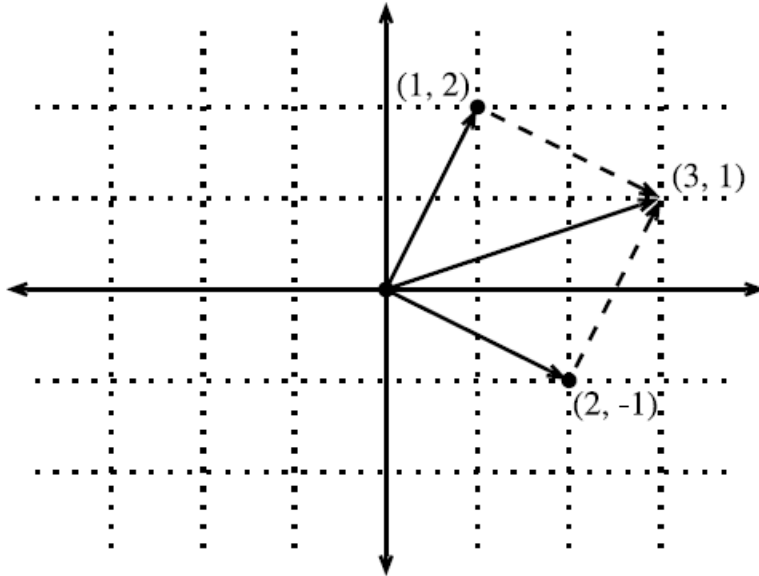
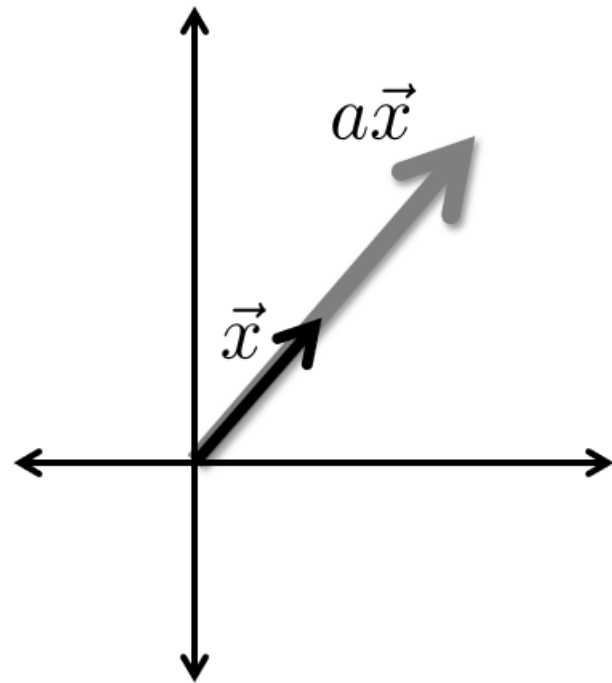
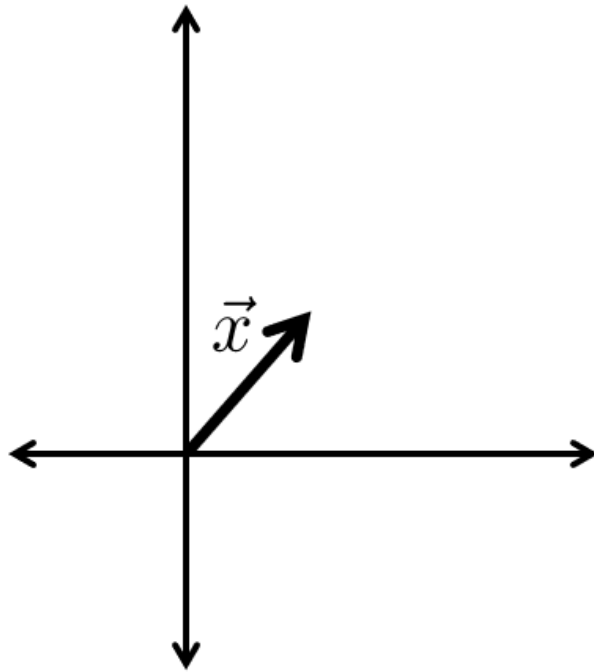


Figure taken from d2l.ai

Geometric view – Scaling a Vector

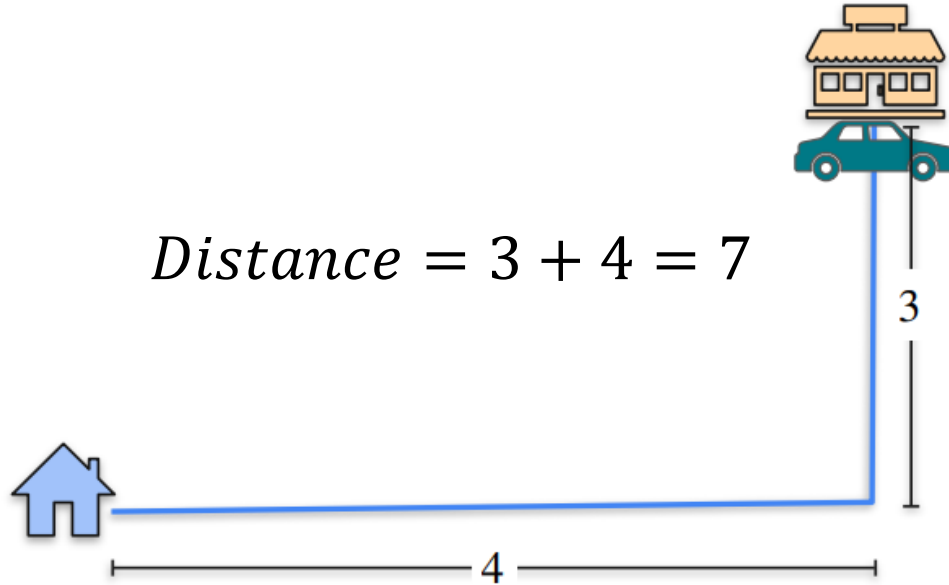


How to get from point A to point B ?



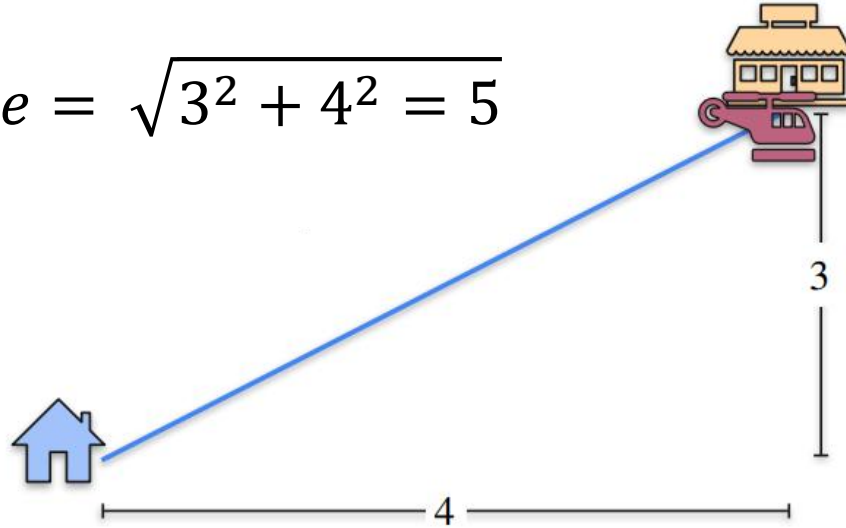
Slide adapted from [DeepLearning.ai](https://deeplearning.ai)

How to get from point A to point B ?



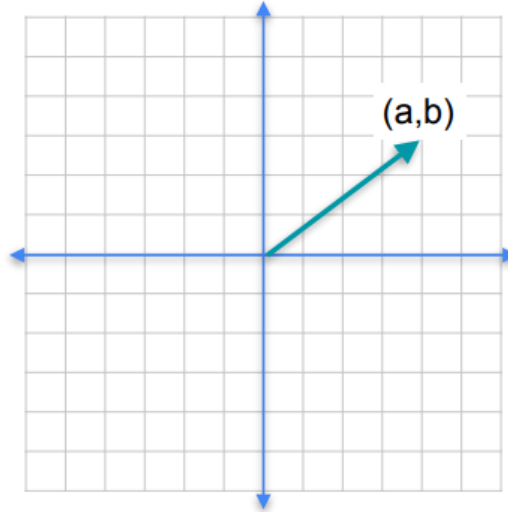
How to get from point A to point B ?

$$\text{Distance} = \sqrt{3^2 + 4^2} = 5$$



Slide adapted from [DeepLearning.ai](https://deeplearning.ai)

Norms



$$\text{L1-norm} = |(a,b)|_1 = |a| + |b|$$



$$\text{L2-norm} = |(a,b)|_2 = \sqrt{a^2 + b^2}$$

Slide adapted from [DeepLearning.ai](https://deeplearning.ai)

Vector Norm

- $\|\mathbf{a}\|_2 = \left[\sum_{i=1}^m a_i^2 \right]^{\frac{1}{2}} = \sqrt{\sum_{i=1}^m a_i^2}$
- $\|\mathbf{a}\|_1 = \left[\sum_{i=1}^m |a_i| \right]$
- $\|\mathbf{a}\| \geq 0$ for all $\mathbf{a}$
- $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$
- $\|a \cdot \mathbf{b}\| = |a| \cdot \|\mathbf{b}\|$

Vector – Vector Multiplication

- Element-wise multiplication
- Inner product (Dot Product)
- Outer product

Element-wise multiplication

- $\mathbf{a} \odot \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \odot \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ a_2 b_2 \\ \vdots \\ a_n b_n \end{pmatrix}$

Inner Product

- $\mathbf{u}^T \mathbf{v} = \sum u_i \cdot v_i$
- $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \mathbf{v}^T \mathbf{u}$

Geometric View – Inner Product

- $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos(\theta)$
- $\theta = \arccos \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right)$

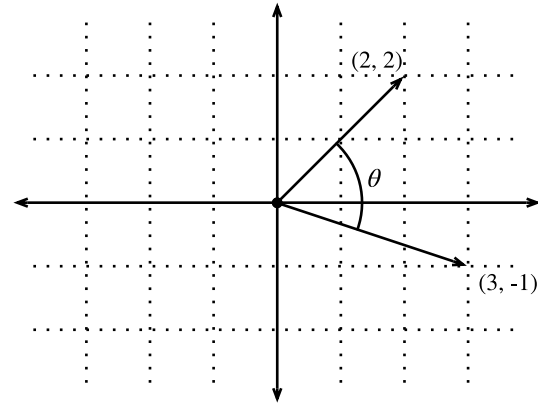
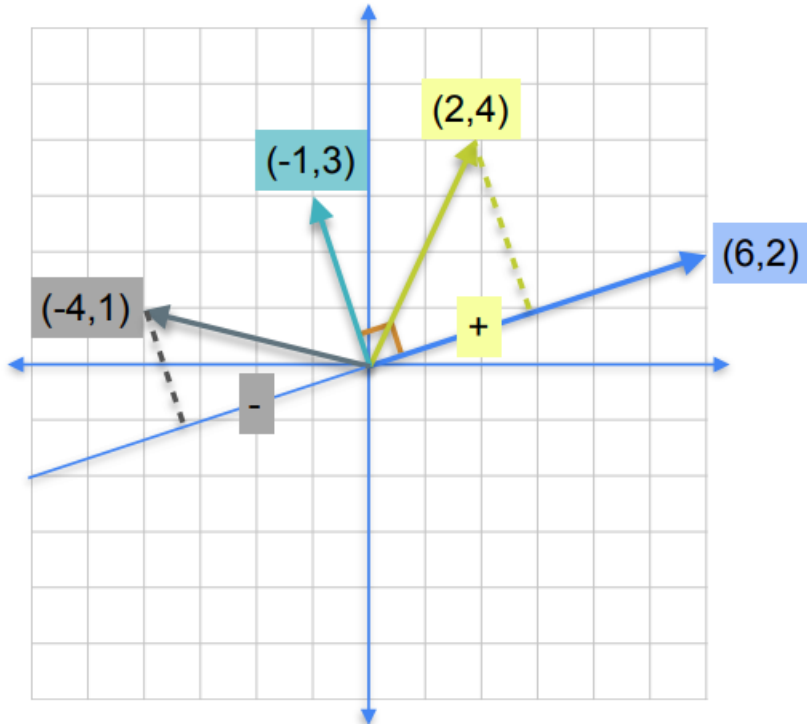


Figure taken from d2l.ai

Visualizing Dot Product



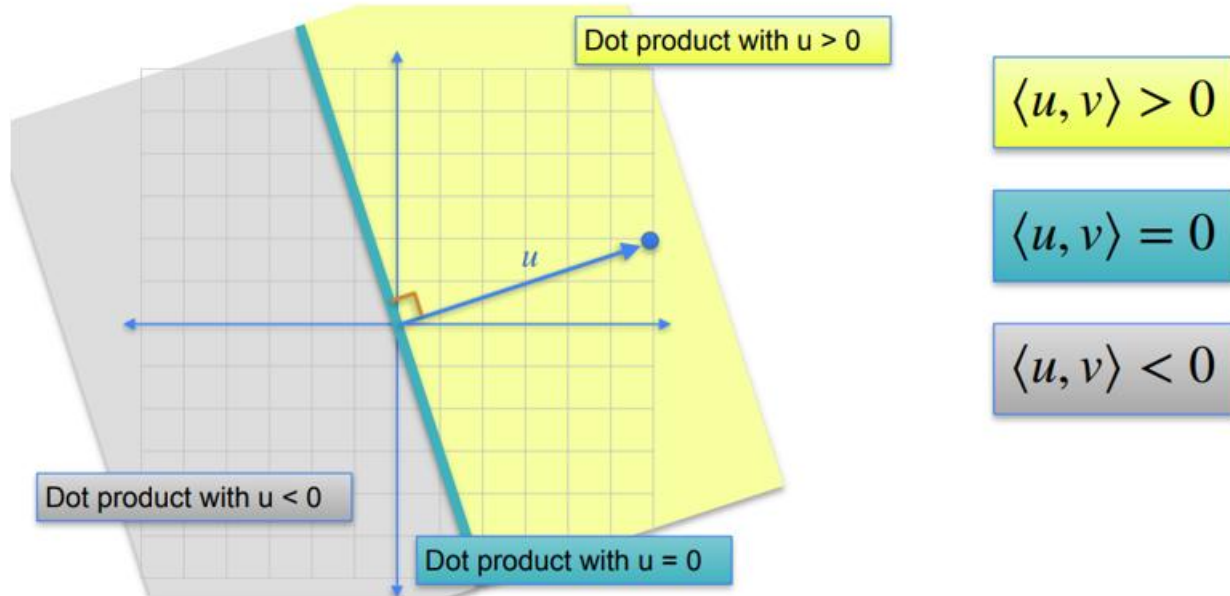
$$\begin{matrix} 6 & 2 \end{matrix} \cdot \begin{matrix} 2 \\ 4 \end{matrix} = 20 \quad \text{Positive}$$

$$\begin{matrix} 6 & 2 \end{matrix} \cdot \begin{matrix} -1 \\ 3 \end{matrix} = 0$$

$$\begin{matrix} 6 & 2 \end{matrix} \cdot \begin{matrix} -4 \\ 1 \end{matrix} = -22 \quad \text{Negative}$$

Slide adapted from [DeepLearning.ai](https://deeplearning.ai)

Visualizing Dot Product



Slide adapted from [DeepLearning.ai](https://deeplearning.ai)

Outer Product

$$\bullet \mathbf{xy}^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \cdot [y_1 \ y_2 \ \cdots \ y_m] = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_m \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_m \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & x_n y_2 & \cdots & x_n y_m \end{bmatrix}$$

Matrix

- Variable described by a list of vectors.
- 2nd order tensor
- Simple Operations
 - Addition
 - Multiplication
 - Function
- Length

Matrix Operations

- Addition

$$\mathbf{C} = \mathbf{A} + \mathbf{B} \text{ where } C_{ij} = A_{ij} + B_{ij}$$

- Scalar Multiplication

$$\mathbf{C} = \alpha \mathbf{B} \text{ where } C_{ij} = \alpha B_{ij}$$

- Function

$$\mathbf{C} = \sin(\mathbf{A}) \text{ where } C_{ij} = \sin(A_{ij})$$

Matrix Norm

- $\|\mathbf{A}\|_{Frobenious} = \left[\sum_{ij} A_{ij}^2 \right]^{\frac{1}{2}}$

Linear Transformation

- $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$
- $\mathbf{A}\mathbf{v} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$
- $\mathbf{A}\mathbf{v} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$
- $\mathbf{A}\mathbf{v} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}$
- $\mathbf{A}\mathbf{v} = x \left\{ \mathbf{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} + y \left\{ \mathbf{A} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

Geometric View – Linear Transformation

- $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

- $v = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

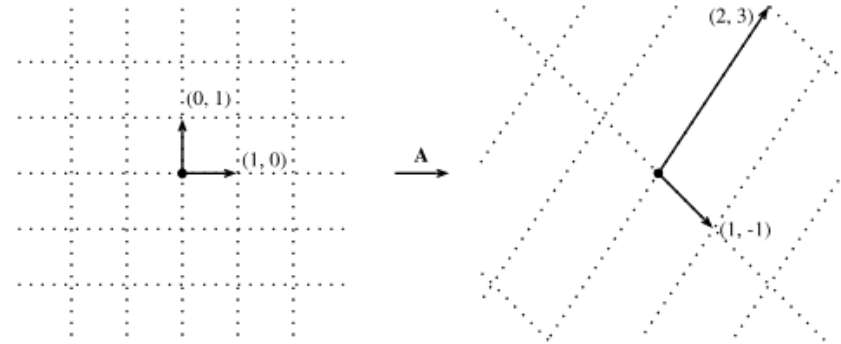


Figure taken from d2l.ai

Matrix – Vector Product

- $A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}$ where each $\mathbf{a}_i^T$ is a row vector representing the i^{th} row of an m by n matrix A .

- $A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{x} \\ \mathbf{a}_2^T \mathbf{x} \\ \vdots \\ \mathbf{a}_m^T \mathbf{x} \end{bmatrix}$ whether i^{th} element is the dot product $\mathbf{a}_i^T \mathbf{x}$

Matrix - Matrix Multiplication

- $A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}$, $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_m]$ where $\mathbf{a}_i^T$ denotes a row of

the matrix A and $\mathbf{b}_j$ denotes a column vector of matrix B .

- $C = AB = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_m] = \begin{bmatrix} \mathbf{a}_1^T \mathbf{b}_1 & \mathbf{a}_1^T \mathbf{b}_2 & \cdots & \mathbf{a}_1^T \mathbf{b}_m \\ \mathbf{a}_2^T \mathbf{b}_1 & \mathbf{a}_2^T \mathbf{b}_2 & \cdots & \mathbf{a}_2^T \mathbf{b}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_n^T \mathbf{b}_1 & \mathbf{a}_n^T \mathbf{b}_2 & \cdots & \mathbf{a}_n^T \mathbf{b}_m \end{bmatrix}$

Identity Matrix

- $I = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$

- *For all A , $AI = IA = A$*

Diagonal Matrix

- $\mathbf{D} = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix}$

- For all $\mathbf{x}$, $\mathbf{D}\mathbf{x} = \begin{bmatrix} d_1 x_1 \\ d_2 x_2 \\ \vdots \\ d_n x_n \end{bmatrix}$

Compressed Notation

- $\boldsymbol{v} \cdot \boldsymbol{w} = \sum_i v_i w_i$
- $\|\boldsymbol{v}\|_2^2 = \sum_i v_i v_i$
- $(\boldsymbol{A}\boldsymbol{v})_i = \sum_j a_{ij} v_j$
- $(\boldsymbol{A}\boldsymbol{B})_{ik} = \sum_j a_{ij} b_{jk}$
- $\text{tr}(\boldsymbol{A}) = \sum_i a_{ii}$

References

- https://d2l.ai/chapter_preliminaries/linear-algebra.html#
- https://d2l.ai/chapter_appendix-mathematics-for-deep-learning/geometry-linear-algebraic-ops.html
- <https://www.deeplearning.ai/courses/mathematics-for-machine-learning-and-data-science-specialization/>